Single-Lung Pneumatic PID Control Design

- Purpose: emulate one 250 mL dynamic lung using a pneumatic circuit, control logic, and UI.

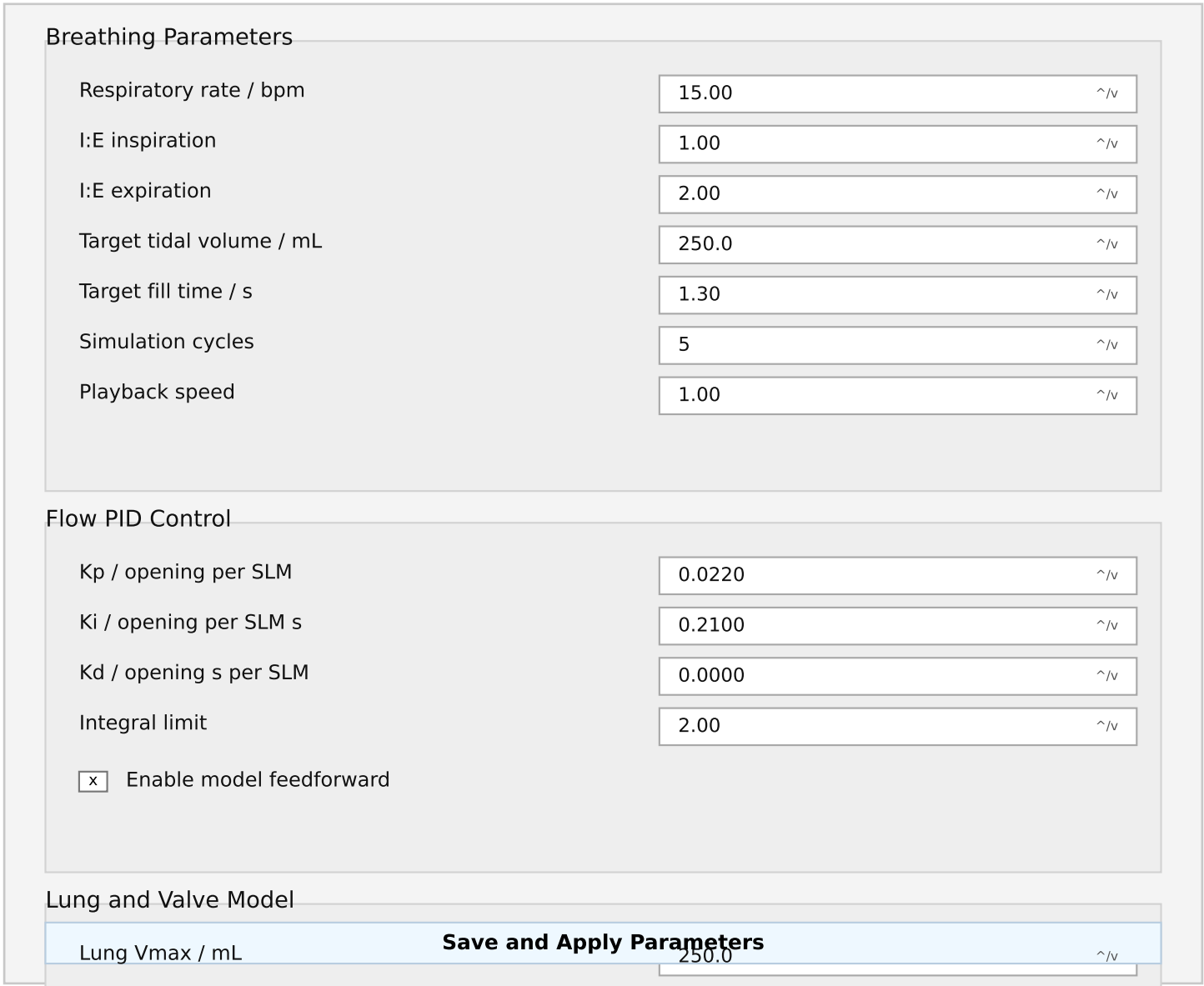
- The UI converts respiratory requirements into valve commands and flow targets.

- Hardware abstraction: one atmosphere solenoid, one vacuum solenoid, and one proportional pinch valve.

- Control objective: fill the lung to the target tidal volume, then exhaust it with repeatable timing.

UI Parameters Used in the Single-Lung Controller

This page reproduces the operator-facing parameters from the PyQt GUI.



How UI Parameters Become Control Actions

The GUI values are converted into timing, flow references, state-machine outputs, and PPV voltage.

Parameter Conversion

- Breathing timing:
RR and I:E calculate inspiration and expiration duration.

- Flow reference:
Target volume and fill time define the desired inflow.

- Valve direction:
The state machine selects ATM open or VAC open.

- Flow regulation:
Kp, Ki, Kd and feedforward compute PPV opening.

- Plant realism:
Vmax, tau, gamma, delay and noise simulate the lung.

- Safety timing:
Dead time closes both solenoids during direction changes.

Example Calculation

RR = 15 bpm	Tcycle = 60 / 15 = 4 s
I:E = 1:2	Tin = 4 x 1/(1+2) = 1.33 s Tout = 4 x 2/(1+2) = 2.67 s
Vtarget = 250 mL, Tfill = 1.30 s	Qin = 0.06 x 250 / 1.30 = 11.54 SLM

Result: the controller knows when to switch states, which solenoid to open, and what flow to track.

Plant Realism Parameters

Model parameters describing lung volume response and pinch-valve nonlinearity.

Parameter	Meaning in the model	Basis / how to determine it
Lung Vmax	Maximum physical lung volume. Here it is set to 250 mL because actual single-lung capacity is 250 mL.	Based on phantom geometry or measured maximum volume.
tau_in	Inspiration time constant. Smaller tau means the lung fills faster for the same valve opening.	Fitted from inflation data: V(t)=Vmax(1-exp(-t/tau_in)).
tau_out	Expiration time constant. Smaller tau means the lung exhausts faster under vacuum.	Fitted from exhaust data: V(t)=V0 exp(-t/tau_out).
gamma	Pinch valve nonlinearity. The model uses: effective opening = opening^gamma.	Fitted from valve calibration: command/opening versus measured flow.

Current tau, gamma, delay and noise values are initial estimates; they should be refined using bench-test data.

Plant Realism Parameters Continued

Model parameters describing sensor realism, safe valve switching, and repeatable initial conditions.

Parameter	Meaning in the model	Basis / how to determine it
transport delay	Delay between a valve command change and the flow sensor response.	Measured from step-response tests. Includes tubing and sensor/filter delay.
flow noise sigma	Standard deviation of random flow noise added to the simulated sensor.	Estimated from steady zero-flow or constant-flow sensor readings.
dead time	Time when both solenoids are closed during direction changes.	Chosen from solenoid response time and safety margin to avoid ATM-VAC cross-connection.
precondition time	Initial exhaust period before formal breathing cycles begin.	Chosen experimentally so each run starts from a consistent low lung volume.

Current tau, gamma, delay and noise values are initial estimates; they should be refined using bench-test data.

From UI Parameters to Breathing Targets

- Respiratory rate defines the cycle period: Tcycle = 60 / RR.

- I:E ratio splits the cycle into inspiration time and expiration time.

- Target tidal volume defines the desired volume change per breath, default 250 mL.

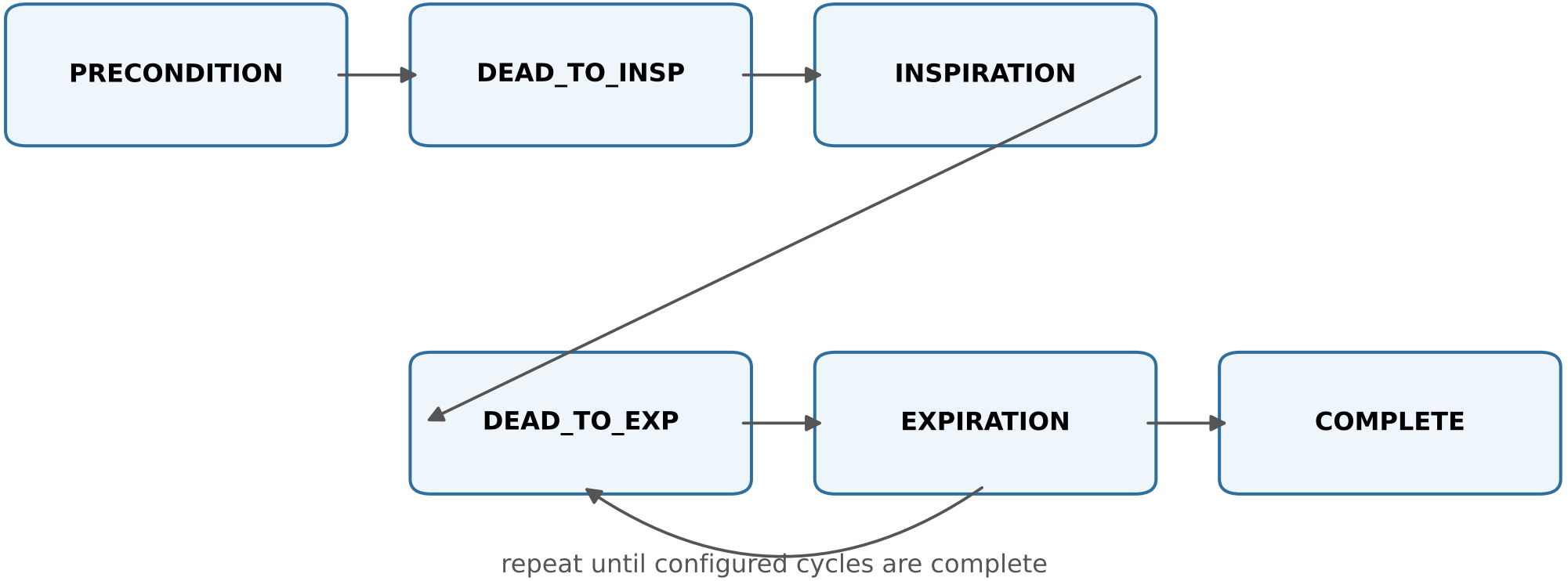
- Target fill time defines the inspiration flow reference: Qin = 0.06 x Vtarget / Tfill.

- Example: Vtarget = 250 mL and Tfill = 1.30 s gives Qin = 11.54 SLM.

- Expiration reference is displayed as negative flow because air is leaving the lung.

State Transition Diagram

- Dead-time states close both solenoids to avoid connecting atmosphere and vacuum simultaneously.
- Inspiration ends by elapsed inspiration time or by reaching the 250 mL target.
- Expiration ends by elapsed expiration time, then the next cycle starts.



State Machine

- PRECONDITION: exhaust the model before the formal breathing cycles start.
- DEAD_TO_INSP: both solenoids closed before switching to inspiration.
- INSPIRATION: atmosphere solenoid open, vacuum solenoid closed, pinch valve modulates inflow.
- DEAD_TO_EXP: both solenoids closed before switching to expiration.
- EXPIRATION: atmosphere solenoid closed, vacuum solenoid open, pinch valve modulates exhaust flow.
- COMPLETE: stop after the configured number of cycles.

Solenoid Valve Simulation

- A compact valve_mask represents the simulated digital outputs.
- bit 0 = atmosphere solenoid command.
- bit 1 = vacuum solenoid command.
- mask 00: both closed, used during dead time and stop states.
- mask 01: atmosphere path open for inspiration.
- mask 10: vacuum path open for expiration/preconditioning.
- This can be mapped directly to Arduino digital output pins later.

Pinch Valve and PID Control

- The solenoid valves choose flow direction; the pinch valve controls flow magnitude.
- Normalized opening a ranges from 0 closed to 1 fully open.
- The valve command is Ucmd = 5(1 - a) V, so 0 V is fully open and 5 V is closed.
- The controller runs at 100 Hz by default and compares target flow magnitude with measured flow magnitude.
- PID law: a = clip(a_ff + Kp e + Ki integral(e dt) + Kd d, 0, 1).
- Feedforward estimates the approximate opening first; PID corrects model and sensor errors.

Single-Lung Plant Model

- The model is a first-order grey-box pneumatic lung model.
- Inspiration: dV/dt = a^gamma (Vmax - V) / tau_in.
- Expiration: dV/dt = -a^gamma V / tau_out.
- Flow conversion: Q = 0.06 dV/dt in SLM.
- Transport delay and sensor noise are added to make the measured flow more realistic.
- The UI displays volume, measured flow, reference flow, pinch opening, command voltage, and valve state.

Summary for Presentation

- The system is a state-machine-based closed-loop pneumatic controller.

- UI parameters define timing and target flow; the state machine selects the pneumatic path.

- Solenoid valves provide direction control: atmosphere for filling, vacuum for exhausting.

- The proportional pinch valve provides continuous flow regulation.

- A feedforward plus PID controller tracks the desired flow and produces a 0-5 V valve command.

- The current software is a simulation/digital twin, but the same valve_mask and command voltage can be mapped to hardware.